SK Dulal Islam

Java Backend / Full-Stack Developer | Spring Boot | Next.js

+91-7364821623 | skdulalislam1@gmail.com | West Bengal, India

LinkedIn | GitHub | Portfolio | LeetCode

Summary

Backend-focused Java developer (final-year B.Tech IT) who builds secure, multi-tenant REST APIs with Spring Boot, Spring Security and PostgreSQL. I care about reliability, performance and clean database design. Looking for a Java Backend or Full-Stack role.

Education

Jalpaiguri Government Engineering College (JGEC) **2023 - 2027**
*B.Tech in Information Technology - **CGPA: 8.0*** *Jalpaiguri, West Bengal*

Mohishageria A M A High Madrasah (HS) **2020 - 2022**
*Higher Secondary (WBCHSE) - **Percentage: 90%*** *Keshpur, West Bengal*

Experience

Internmo **July 2026 - Present**
Full Stack Developer Intern *Remote*

MedCore HMS - Multi-Tenant Hospital Management System (Spring Boot, PostgreSQL, Next.js)

- Built authentication with **Spring Security and JWT** (refresh token rotation) and **role-based access control** for 7 user roles.
- Implemented **multi-tenant data isolation**: every request resolves the current hospital and all queries are scoped to it.
- Designed reliable invitation emails with the **Outbox pattern**, **retry with exponential backoff and jitter**, and **idempotency keys** backed by a database unique constraint to stop duplicate requests.
- Used **@Transactional** with **integration tests** on a separate PostgreSQL database to verify rollback and concurrency.
- Used **EXPLAIN ANALYZE**, added **indexes** via **Flyway** migrations, and fixed **N+1 queries** with `@EntityGraph`.
- Added **Redis caching** with cache invalidation on updates, and **API rate limiting**, to reduce database load.
- Owned the **Billing module** end to end (schema, services, REST APIs) with DTOs, validation, pagination and Swagger.
- Built the **Next.js** frontend; set up **Docker Compose, Nginx** and **GitHub Actions CI/CD**.

Projects

ShopLy - Multi-Vendor E-Commerce Platform (Spring Boot, React, Redux Toolkit, MySQL, JWT)

- Built Customer, Seller and Admin modules with JWT and RBAC, seller approval flow, and REST APIs for products, cart, checkout and orders; integrated **Razorpay** and **Cloudinary**.

EduZen - Coaching Centre Management System (Spring Boot, React, MySQL, JWT)

- Built Admin and Student modules with Spring Security, JWT, fee tracking and approval workflows.

URL Shortener (Java 21, Spring Boot, Spring Security, JWT, MySQL)

- Built a secure backend with JWT and refresh tokens, ownership-based authorization, custom aliases, click tracking and link expiry.

Technical Skills

- **Languages and CS:** Java, SQL, JavaScript, TypeScript, C++, Data Structures and Algorithms, OOP, DBMS
- **Backend:** Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, JWT, RBAC, Swagger/OpenAPI, JUnit 5, Node.js
- **Databases and Caching:** PostgreSQL, MySQL, MongoDB, Redis, Flyway, Database Indexing
- **Concepts:** Multi-Tenancy, Transactions, Idempotency, Outbox Pattern, Caching, Rate Limiting, N+1 Optimization
- **Frontend:** Next.js, React.js, Redux Toolkit, Tailwind CSS, Bootstrap
- **DevOps and Tools:** Git, GitHub, Docker, Docker Compose, Nginx, GitHub Actions, Maven, Postman, Render, Vercel
- **Currently Learning:** System Design

Achievements

- Secured **5th place** in CSS Battle at Sristi 2025, JGEC's annual techno-cultural fest.

Extracurricular

Coders Club DnC - JGEC **2023 - Present**
Active Member

- Participate in coding contests, peer code reviews and technical workshops; contribute to a collaborative problem-solving culture.